Final Report

Risk Factors Associated with Low Infant Birth Weight

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An Analysis of Risk Factors Associated with Low Infant Birth Weight

Abstract

This study investigates the factors associated with low infant birth weight (LBW) and evaluates the effectiveness of predictive models using the Baystate Medical Center Birth dataset. Logistic Regression, optimized at a threshold of 0.34, emerged as the most effective model for predicting LBW, achieving an F1-score of 0.64 and recall of 0.67. Key predictors identified include last menstrual weight, maternal smoking, and a history of hypertension, emphasizing the importance of maternal health and behaviors. The findings provide critical insights into addressing LBW and improving neonatal outcomes through targeted interventions and public health strategies.

1. Data Origin and Overview

The dataset was obtained from the Baystate Medical Center Birth database. It consists of 11 variables and 189 observations, each representing a birth record. Key variables include the infant's birth weight, maternal age, smoking status, hypertension history, and last menstrual weight. The dataset provides an opportunity to explore relationships between maternal health, behaviors, and demographic factors in predicting low birth weight, defined as a weight of less than 2,500 grams. The primary goal was to uncover significant predictors of LBW and evaluate predictive modeling approaches for accurate identification of at-risk cases.

2. Data Preprocessing

2.1 Data Cleaning

In the initial stages of data preprocessing, variable names were renamed to enhance readability. The birth_weight_grams column was removed as it directly contributed to generating the boolean target variable, low_birth_weight. Similarly, the weight column was excluded due to its unclear relevance to the analysis. Duplicate records were identified and a total of six duplicates were removed to maintain dataset integrity.

2.2 Missing Values

The dataset did not contain missing values, allowing for direct analysis without the need for imputation techniques.

2.3 Extreme Outliers

Extreme outliers were assessed using the interquartile range (IQR) method. A few outliers in the last menstrual weight variable were identified but retained, to preserve a comprehensive data set, given small size of our data set, and the fact that they were not too extreme and therefore did not affect our classification model performance.

2.4 Data Mutation

Data mutation was employed to convert the numerical age_of_mother variable into the categorical age_category, segmenting ages into "Teen," "Young Adult," and "Mature Adult" groups for nuanced analysis. The low_birth_weight variable was also transformed into a boolean data type to clearly denote the binary nature of this target variable.

These data mutation steps were not just beneficial but necessary to optimize the analytical process and derive actionable insights. They allowed the analysis to align more closely with the research objectives by facilitating comparisons across key demographic segments and enabling robust predictive modeling. Additionally, these transformations provided a clearer narrative around the dataset’s structure and the phenomena being studied, ensuring that both human and machine-driven analyses could accurately capture the nuances of the data.

2.5 One-hot Encoding

Categorical variables were converted to boolean datatypes using one-hot encoding this enables machine learning models to effectively process and analyze the data.

3. Exploratory Data Analysis

3.1 Numerical Variables

The dataset includes two numerical variables: the age of the mother and the last menstrual weight. The average age of the mothers is 23.42 years, with a median of 23 years. The ages range from 14 to 45, and most mothers fall within the 20 to 26.5-year age range. The distribution of ages appears to be fairly symmetrical, with the histogram showing a relatively uniform spread, suggesting that younger mothers are more common in the dataset. Regarding last menstrual weight, the average weight is 130.09 pounds, with a median of 121 pounds and a standard deviation of 30.86 pounds. Weights range from 80 pounds to 250 pounds, and the histogram reveals a slight right skew, indicating that while most mothers have lower weights, there are a few extreme higher weights in the dataset. Overall, these variables provide insights into the age and weight characteristics of the mothers, with a notable concentration of younger mothers and a wider variation in weight.

3.2 Categorical Variables

The dataset includes several categorical variables that provide important context regarding maternal and infant health. The low_birth_weight variable indicates whether an infant's birth weight was below 2,500 grams. Of the 183 infants, 68.3%did not have low birth weight, while 32.2% did, highlighting the proportion of infants with a birth weight that could pose potential health risks. The smoking_status variable reveals that 38.3% mothers smoked during pregnancy, This significant proportion of mothers who smoked is particularly relevant, as smoking during pregnancy is known to negatively impact birth outcomes, including low birth weight. The mother_race variable shows that 50.3% of mothers are White, 36.1% are Other, and 13.7% \are Black. This distribution could help in understanding the role of racial or ethnic factors in maternal and infant health outcomes. The age_category variable classifies mothers into three groups: 50.3% are Young Adults, 24.6% are Teens, and 25.1% (are Mature Adults. This categorization is important for analyzing age-related patterns in maternal health, as the age of the mother can influence pregnancy outcomes. The dataset also includes several health-related variables. For hypertension_history, only 6.6% mothers reported a history of hypertension out of 183, suggesting that hypertension was not a common issue in this cohort. Uterine_irritability was reported by 14.8% mothers, and premature_labors were reported by 16.4% mothers, both of which are factors that can contribute to complications during pregnancy and birth. In terms of prenatal care, physician_visit indicates that 48.1% mothers had a physician visit during the first trimester, while 51.9% did not, implying that nearly half of the mothers in the dataset did not receive early prenatal care, which could be a significant factor in maternal and infant health outcomes.

Together, these categorical variables provide valuable information on the socio-demographic and health conditions of the mothers in the dataset. They offer important context for understanding the potential predictors of low birth weight and other maternal health outcomes. The final analysis will include visualizations to help further clarify the relationships between these variables.

3.3 Variable Combinations

A bivariate analysis was next conducted to explore the relationship between smoking status (smoking_status) and low birth weight (low_birth_weight). After normalizing the data to account for differences in group sizes, it became evident that mothers who smoked during pregnancy were significantly more likely to have a baby with low birth weight compared to non-smoking mothers. Specifically, 42.86% of smoking mothers had babies with low birth weight, compared to only 25.66% of non-smoking mothers.This disparity is further highlighted in the countplot (A.1), where the proportion of low birth weight outcomes is visibly higher for smoking mothers. These findings suggest a strong association between maternal smoking and the likelihood of low birth weight, reinforcing the importance of addressing smoking behaviors during pregnancy to improve neonatal health outcomes.

In exploring the relationship between last menstrual weight and low birth weight (LBW), a bivariate analysis was conducted to assess whether there was a noticeable trend between these two variables. The findings suggest that mothers who gave birth to infants with low birth weight generally had lower last menstrual weights compared to those who gave birth to infants with normal birth weight.This relationship was visualized using a box plot, which clearly distinguishes the distribution of last menstrual weights for both the LBW and non-LBW groups. The box plot illustrates the median, mean, and the spread of data for each group, highlighting that mothers with LBW infants tend to have lower weights, as seen by the lower mean and median values in the LBW group. The non-LBW group, on the other hand, shows a higher mean and median, suggesting that greater weight at the time of the last menstrual period may be associated with more favorable birth outcomes.

This trend is significant as it emphasizes the potential role of maternal weight, specifically at the time of the last menstrual period, as an indicator of pregnancy outcomes. The box plot further reveals the variation within each group, illustrating that while the general trend holds, there is still considerable individual variation. These insights suggest that last menstrual weight could serve as a valuable predictor for assessing the likelihood of low birth weight, although further research is needed to explore other contributing factors and strengthen this relationship.

4. Model selection and Comparison of Supervised Machine Learning Models.

Logistic Regression was the best-performing model, achieving an F1-score of 0.64 and recall of 0.67 at an optimal threshold of 0.34. This threshold effectively balanced precision and recall, minimizing false negatives and identifying at-risk cases more reliably.

Comparison of Supervised Machine Learning Models

Four supervised machine learning models were evaluated for their ability to predict LBW: Logistic Regression, Decision Tree, Tuned Decision Tree, and K-Nearest Neighbor (KNN). Each model was trained and tested on a split dataset, with performance metrics such as accuracy, F1-score, and recall serving as evaluation criteria.

Logistic Regression with an optimized threshold of 0.34 emerged as the best-performing model. It achieved an accuracy of 76%, an F1-score of 0.64, and a recall of 0.67. These metrics indicate a strong balance between precision and recall, particularly in minimizing false negatives. By comparison, the standard Logistic Regression model without threshold optimization achieved lower recall, underscoring the value of threshold tuning.

The Decision Tree model demonstrated poor overall performance, with an accuracy of 54% and an F1-score of 0.32. Although the Tuned Decision Tree improved recall to 67%, its accuracy dropped to 49%, reflecting a trade-off between recall and precision. Finally, the KNN model achieved high accuracy (68%) but failed to detect true LBW cases effectively, as indicated by its recall of only 8%.

Summary of Model Performance

Logistic Regression outperformed other models, providing a reliable framework for predicting LBW outcomes.

5. Summary of Findings

The analysis of the Baystate Medical Center Birth dataset provided several important insights into the factors associated with low birth weight (LBW) and the effectiveness of predictive modeling. Below are the updated findings:

Predictive Modeling:

Logistic Regression, optimized with an F1-score of 0.64 and a recall of 0.67 at a threshold of 0.34, was identified as the most effective model for predicting low birth weight. This model balanced precision and recall, effectively minimizing false negatives.

Key Predictors of Low Birth Weight:

Last Menstrual Weight: The most significant predictor of LBW, with an importance score of 0.705. Mothers with lower last menstrual weights were strongly associated with having babies with LBW, emphasizing its critical role in understanding neonatal outcomes.

Smoking During Pregnancy: Smoking was another key predictor, with 42.86% of smoking mothers having LBW infants compared to 25.66% of non-smoking mothers. This highlights the impact of maternal behavior on birth outcomes.

History of Hypertension: A history of hypertension significantly contributed to LBW predictions, with an importance score of 0.172, underlining the role of maternal health conditions.

Maternal and Demographic Insights:

The analysis showed disparities in maternal health factors.

Younger mothers and those with lower last menstrual weights or specific health conditions (e.g., hypertension) were more likely to have LBW infants.

Behavioral factors like smoking were strongly associated with LBW outcomes, reinforcing the need for behavioral health interventions during pregnancy.

6. Implications and Recommendations

There were some recommendations, based on the research and findings. It is important to ensure that expecting mothers maintain healthy weights through counseling on balanced diets and the provision of nutritional supplements, with regular weight monitoring integrated into prenatal visits. Additionally, robust protocols should be developed for identifying and managing hypertension in pregnant women, which should include lifestyle counseling, medications, and stress reduction techniques. Smoking cessation programs tailored to pregnant women should also be implemented, combining counseling, support groups, and nicotine replacement therapies. Furthermore, public awareness campaigns should be launched to educate about the risks of smoking and hypertension during pregnancy, aiming to promote healthier behaviors

7. Conclusion

This study underscores the importance of maternal health and behavioral factors in predicting low birth weight, emphasizing actionable strategies to mitigate these risks. Logistic Regression proved to be the most effective predictive model, identifying significant predictors such as last menstrual weight, smoking, and hypertension.

Birth weight remains a critical concern due to its association with infant mortality and birth defects. The World Health Organization (WHO) defines low birth weight as less than 2,500 grams, a threshold that highlights its clinical importance. Maternal behavior during pregnancy, such as smoking cessation and managing hypertension, plays a pivotal role in improving outcomes.

The motivation behind this research was to identify and address these modifiable risk factors, providing actionable insights for improving neonatal outcomes. By leveraging data analytics and predictive modeling, healthcare professionals can implement targeted interventions to support at-risk mothers and reduce the prevalence of LBW.

Project Limitations

The dataset exhibited an imbalanced distribution of low birth weight (LBW) cases, which may have impacted the performance of the model and the insights derived from it. Additionally, the relatively small sample size of 189 cases posed a limitation, restricting the generalizability of the findings and potentially reducing the robustness of the conclusions drawn from the analysis.

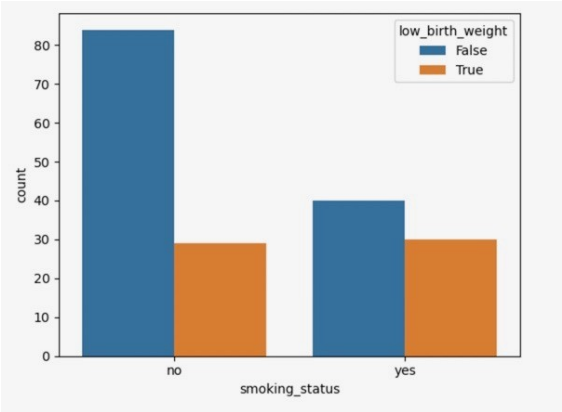
Future Work

To enhance the analysis, it is important to expand data collection by gathering more extensive datasets over longer timeframes. This will help identify more robust patterns and predictors, providing a clearer picture of the factors involved. Additionally, attention should be given to data balance by incorporating strategies to handle imbalanced datasets. Techniques such as oversampling or synthetic data generation can be employed to ensure improved model training and more accurate outcomes. Furthermore, the analysis could benefit from incorporating additional variables. By exploring other potential predictors, including socioeconomic and environmental factors, a more comprehensive understanding of the subject can be achieved, offering deeper insights into the risks associated with low infant birth weight.

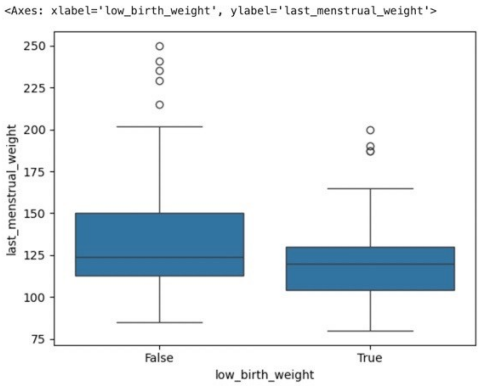
8. References

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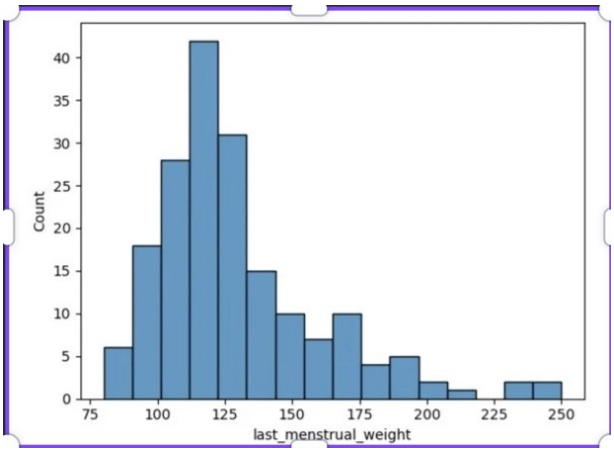
9. Appendix



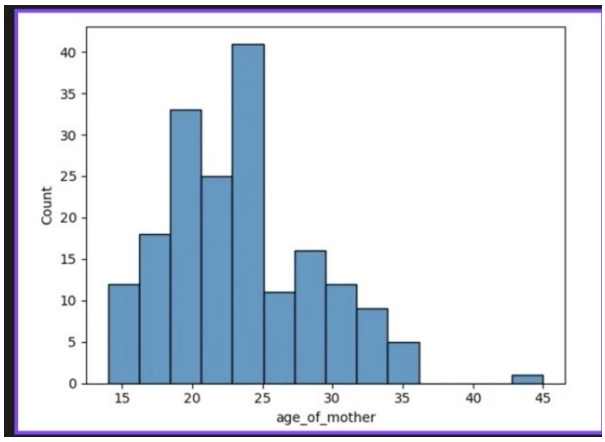
A.1 ***Histogram showing impact of smoking on LBW***



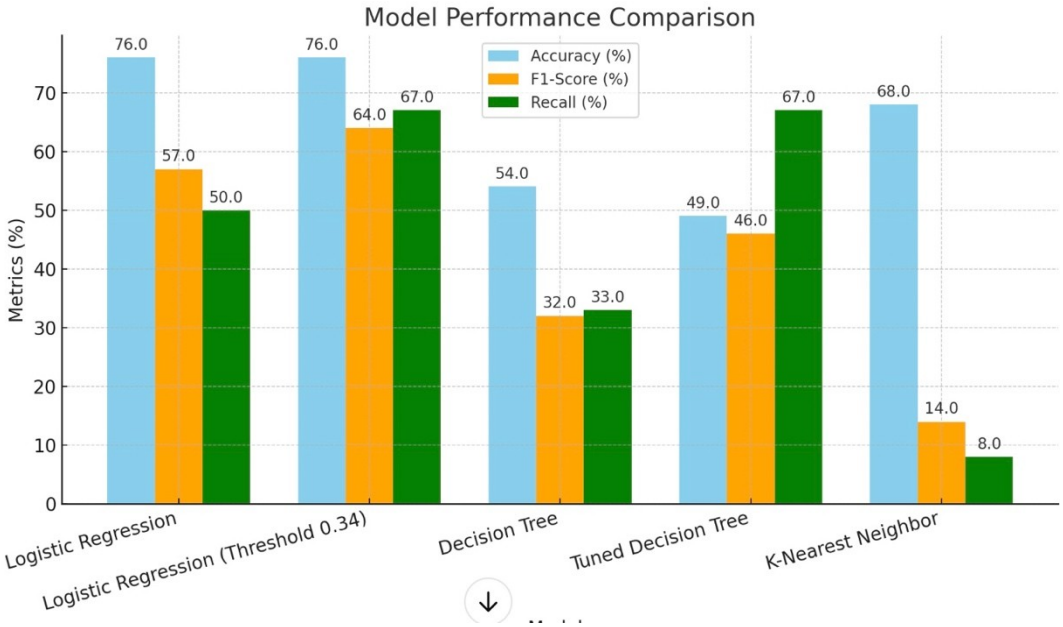
A.2 Box plot showing Last Menstrual Weight and Low Birth Weight



A.3.histogram showing last_menstrual_weight distribution



A.3.histogram showing age_of_mother distribution



A.5.Model performance comparison histogram